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SEMICONDUCTOR STRUCTURE AND METHOD FOR MANUFACTURING A SEMICONDUCTOR STRUCTURE

Abstract

The present application discloses a semiconductor package and a method for manufacturing the semiconductor package. The semiconductor package includes a first dielectric, a first redistribution layer (RDL) disposed on a first surface of the first dielectric, a first bonding layer disposed on the first RDL, a plurality of bottom dies attached to the first bonding layer, a second dielectric filling gaps between the bottom dies, a plurality of conductive pillars disposed in the second dielectric without contacting the bottom dies, a second RDL disposed on the second dielectric and the bottom dies, a second bonding layer disposed on the second RDL, a plurality of top dies attached to the second bonding layer, a third dielectric filling gaps between the top dies, and a plurality of solder bumps disposed on a second surface of the first dielectric.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a semiconductor structure, and more particularly, to a semiconductor structure having a power management integrated circuit stacking on silicon capacitor layers.

BACKGROUND

[0002] In electronic devices, different components and modules often require different voltages to operate. Therefore, electronic devices need to convert input voltages, such as a battery voltage or a USB power supply voltage, into different voltage rails to meet the needs of various components and modules. To facilitate power management, electronic devices often adopt power management integrated circuits (PMIC) to perform voltage conversion and provide multiple voltages required by the system.

[0003] Voltage conversion circuits in the PMIC often require passive components, such as capacitors, to store energy. The quality and quantity of these passive components can significantly impact the power efficiency of conversion process. However, due to the large area occupied by passive components, electronic devices with limited space may face trade-off between area and power efficiency. Conventionally, a PMIC and capacitors are typically placed on a same substrate and connected through conductive traces within the substrate. In such case, the conductive paths between the PMIC and the capacitors can be rather long and may cross different material interfaces, which results in poor signal integrity and power integrity. Therefore, how to improve the efficiency of voltage conversion remains a problem to be solved.

SUMMARY

[0004] One aspect of the present disclosure provides a semiconductor structure. The semiconductor structure includes a first chip, a second chip, a first bonding layer, and a power management die. The first chip has a first 3-dimensional (3D) capacitor structure therein, and the second chip has a second 3D capacitor structure therein. The first bonding layer bonds the first chip with the second chip. The power management die is bonded with the second chip. The power management die is electrically connected to the first 3D capacitor structure and the second 3D capacitor structure through interconnects extending along a thickness of the semiconductor structure and across the first bonding layer.

[0005] Another aspect of the present disclosure provides a semiconductor device. The semiconductor device includes a printed circuit board (PCB), a substrate, a system on chip (SoC), and the semiconductor structure aforementioned. The substrate is disposed on the PCB, and the SoC is disposed on a first surface of the substrate. The semiconductor structure is coupled to the SoC.

[0006] Another aspect of the present disclosure provides a method for manufacturing a semiconductor structure. The method includes forming a bonding between a first wafer having a plurality of first 3D capacitor structures therein and a second wafer having a plurality of second 3D capacitor structures therein to bond the first wafer with the second wafer, bonding a plurality of power management dies over the second wafer, and performing a sawing process to form the semiconductor structure including a first chip, a second chip, a first bonding layer, one of the plurality of power management dies, one of the plurality of first 3D capacitor structures in the first

chip and one of the plurality of second 3D capacitor structures in the second chip. The power management die is electrically connected to the first 3D capacitor structure and the second 3D capacitor structure through interconnects extending along a thickness of the semiconductor structure and across the first bonding layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more complete understanding of the present disclosure may be derived by referring to the detailed description and claims when considered in connection with the Figures, where like reference numbers refer to similar elements throughout the Figures.

[0008] FIG. 1 shows a semiconductor structure according to one embodiment of the present disclosure.

[0009] FIG. 2 shows a flow chart of a method for manufacturing the semiconductor structure in FIG. 1 according to one embodiment of the present disclosure.

[0010] FIGS. 3A to 3K show cross-sectional view or side view of one or more stages for manufacturing the semiconductor structure in FIG. 1 according to one embodiment of the present disclosure.

[0011] FIG. 4 shows a semiconductor structure according to one embodiment of the present disclosure.

[0012] FIGS. 5A to 5H show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 4 according to one embodiment of the present disclosure.

[0013] FIG. 6 shows a semiconductor structure according to one embodiment of the present disclosure.

[0014] FIGS. 7A to 7H show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 6 according to one embodiment of the present disclosure.

[0015] FIG. 8 shows a semiconductor structure according to one embodiment of the present disclosure.

[0016] FIGS. 9A to 9D show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 8 according to one embodiment of the present disclosure.

[0017] FIGS. 10A to 10E show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 6 according to one embodiment of the present disclosure.

[0018] FIG. 11 shows a semiconductor structure according to one embodiment of the present disclosure.

[0019] FIGS. 12A to 12K show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 11 according to one embodiment of the present disclosure.

[0020] FIG. 13 shows a semiconductor structure according to one embodiment of the present disclosure.

[0021] FIGS. 14A to 14H show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 13 according to one embodiment of the present disclosure.

[0022] FIGS. 15A to 15E show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 13 according to one embodiment of the present disclosure.

[0023] FIG. 16 shows a semiconductor structure according to one embodiment of the present disclosure.

[0024] FIGS. 17A to 17E show cross-sectional view of one or more stages for manufacturing the semiconductor structure in FIG. 16 according to one embodiment of the present disclosure.

[0025] FIGS. 18 to 22 show semiconductor devices according to embodiments of the present disclosure.

[0026] FIG. 23 shows a cross-sectional view of a 3D capacitor structure according to some

embodiments of the present disclosure.

[0027] FIG. **24** shows a cross-sectional view of a 3D capacitor structure according to some embodiments of the present disclosure.

[0028] FIG. **25A** shows a top view of a rectangular-array-arrangement of 3D capacitor unit cells according to some embodiments of the present disclosure.

[0029] FIG. **25B** shows a top view of a hexagonal-array-arrangement of 3D capacitor unit cells according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0030] The following description accompanies drawings, which are incorporated in and constitute a part of this specification, and which illustrate embodiments of the disclosure, but the disclosure is not limited to the embodiments. In addition, the following embodiments can be properly integrated to complete another embodiment.

[0031] References to “one embodiment,” “an embodiment,” “exemplary embodiment,” “other embodiments,” “another embodiment,” etc. indicate that the embodiment(s) of the disclosure so described may include a particular feature, structure, or characteristic, but not every embodiment necessarily includes the particular feature, structure, or characteristic. Further, repeated use of the phrase “in the embodiment” does not necessarily refer to the same embodiment, although it may.

[0032] In order to make the present disclosure completely comprehensible, detailed steps and structures are provided in the following description. Obviously, implementation of the present disclosure does not limit special details known by persons skilled in the art. In addition, known structures and steps are not described in detail, so as not to unnecessarily limit the present disclosure. Preferred embodiments of the present disclosure will be described below in detail. However, in addition to the detailed description, the present disclosure may also be widely implemented in other embodiments. The scope of the present disclosure is not limited to the detailed description, and is defined by the claims.

[0033] FIG. **1** shows a semiconductor structure **100** according to one embodiment of the present disclosure. The semiconductor structure **100** includes a chip **110**, a chip **120**, a power management die **130**, and a bonding layer **140**. The bonding layer **140** bonds the chip **110** with the chip **120**, and the power management die **130** is bonded with the chip **120**. In the present embodiment, the semiconductor structure **100** further includes a bonding layer **150**, and the power management die **130** is bonded with the chip **120** by the bonding layer **150**.

[0034] In some embodiments, the chip **110** and the chip **120** can be capacitor chips that each includes at least one 3D capacitor structure formed therein. For example, the chip **110** includes at least one 3D capacitor structure **C1**, the chip **120** includes at least one 3D capacitor structure **C2**. In some embodiments, the chip **110** and the chip **120** may be identical or having similar layouts. Thus, the 3D capacitor structure **C1** may be aligned with the 3D capacitor structure **C2**. In some embodiments, the 3D capacitor structure **C1** is vertically aligned with the 3D capacitor structure **C2**. In some embodiments, from a top view of the semiconductors structure **100**, the 3D capacitor structure **C1** overlaps with the 3D capacitor structure **C2**. However, the present disclosure is not limited thereto.

[0035] The power management die **130** can be a logic die that includes a power management integrated circuit (PMIC). The PMIC can be adopted to convert a supply voltage into different voltage rails required by components in a system. In the present embodiment, the power management die **130** can be coupled to the capacitors, such as the 3D capacitor structures **C1** and **C2**, in the chip **110** and the chip **120** for voltage conversion. In such case, the power management die **130** is electrically connected to the 3D capacitor structure **C1** and the 3D capacitor structure **C2** at least through interconnects extending along a thickness (as shown by an arrow in FIG. **1**) of the semiconductor structure **100**. That is, the stacked structure of the chip **110**, the chip **120** and the power management die **130** allows the power management die **130** to couple to the 3D capacitor structure **C1** formed in the chip **110** through conductive paths crossing the bonding layer **140** and

the bonding layer **150** and to couple to the 3D capacitor structure **C2** formed in the chip **120** through conductive paths crossing the bonding layer **150**. As a result, the power management die **130** can be coupled to the capacitance provided by the chips **110** and **120** in a smaller area with shorter routing distances, thereby improving the power efficiency, the signal integrity, and the power integrity.

[0036] As shown in FIG. **1**, the chip **110** may include a semiconductive layer **112**, a dielectric **114**, at least one via **116**, and a redistribution layer (RDL) **118**. In some embodiments, the semiconductive layer **112** may include semiconductive material such as silicon, germanium, gallium, arsenic, or a combination thereof. In some embodiments, the semiconductive layer **112** can be a wafer substrate. In some embodiments, the semiconductive layer **112** is a silicon substrate. In some embodiments, the semiconductive layer **112** includes a first surface **112a** and a second surface **112b** opposite to the first surface **112a**. The first surface **112a** is in proximity to dielectric **114**, and the second surface **112b** is farther away from the dielectric **114**. In some embodiments, the via **116** extends through the semiconductive layer **112** from the first surface **112a** to the second surface **112b**. In some embodiments, the semiconductive layer **112** is a silicon substrate, and the via **116** is a through silicon via (TSV).

[0037] The dielectric **114** is disposed over the semiconductive layer **112**. In some embodiments, the dielectric **114** is disposed on the first surface **112a** of the semiconductive layer **112**. In some embodiments, the dielectric **114** includes dielectric material, such as silicon oxide, silicon nitride, polymer, or the like. In some embodiments, the dielectric **114** includes several dielectrics layers stacking over each other. In some embodiments, the dielectrics includes same material or different materials. In some embodiments, the dielectric **114** is interlayer dielectric (ILD).

[0038] In some embodiments, at least one interconnect structure **114a** and the at least one 3D capacitor structure **C1** are formed within the dielectric **114**. In some embodiments, the interconnect structure **114a** is an electrical routing within the chip **110**. For example, the interconnect structure **114a** may include metal lines that extend laterally at different levels and metal vias that extend vertically for connecting metal lines at different levels. In some embodiments, the interconnect structure **114a** is formed in a back end of line (BEOL) process.

[0039] As shown in FIG. **1**, the chip **110** further includes a plurality of conductive bumps **119**. The conductive bumps **119** can be solder balls or copper pillars that are coupled to the interconnect structure **114a** formed in the dielectric **114** for external connections of the semiconductor structure **100**. In some embodiments, the conductive bumps **119** may include conductive material, such as copper, tin, silver, lead, alloy of the aforementioned materials, or the like.

[0040] The chip **110** has a first surface **110a** and a second surface **110b** opposite to the first surface **110a**. In some embodiments, the first surface **110a** is in proximity to dielectric **114** and is referred as a front side of the chip **110**, and the second surface **110b** is in proximity to semiconductive layer **112** with the via **116** and is referred as a back side of the chip **110**.

[0041] As shown in FIG. **1**, the conductive bumps **119** are disposed on the first surface **110a**. In addition, the bonding layer **140** can contact the RDL **118** at the second surface **110b** of the chip **110**. In some embodiments, the bonding layer **140** can be formed by bonding a sub-bonding layer **140a** with a sub-bonding layer **140b**.

[0042] The sub-bonding layer **140a** includes a bonding dielectric **141a** and at least one bonding pad **142a**. In some embodiments, the bonding pad **142a** is surrounded by and at least partially exposed through the bonding dielectric **141a**. In some embodiments, to facilitate the connection between the via **116** in the semiconductive layer **112** and the first bonding pads **142a** in the sub-bonding layer **140a**, the RDL **118** is disposed over the semiconductive layer **112** and is coupled to the via **116** and the bonding pad **142a**. Although FIG. **1** only shows one layer of metal in the RDL **118**, the present disclosure is not limited thereto. In some other embodiments, the RDL **118** may include more layers of metals with interlayer dielectrics formed there in for separating the different layers of metals.

[0043] In some embodiments, the bonding pad **142a** extends through the bonding dielectric **141a** to electrically connect to the metal lines in the RDL **118**. Thus, the bonding pad **142a** can be electrically connected to the first 3D capacitor structure **C1** through the RDL **118**, the via **116**, and the interconnect structure **114a**. In some embodiments, the bonding pad **142a** includes conductive material such as copper, silver or the like.

[0044] In some embodiments, the chip **120** and the chip **110** may have similar structures. For example, the chip **120** includes a dielectric **124**, a semiconductive layer **122** over the dielectric **124**, at least one via **126** extending through the semiconductive layer **122**, and a RDL **128** over the semiconductive layer **122**. As shown in FIG. **1**, the chip **120** has a first surface **120a** and a second surface **120b** opposite to the first surface **120a**. In some embodiments, the first surface **120a** is referred as a front side of the chip **120**, and the second surface **120b** is referred as a back side of the chip **120**. However, unlike the chip **110**, instead of having conductive bumps **119** disposed on the first surface **110a**, the sub-bonding layer **140b** is disposed on the first surface **120a** of the chip **120**. The sub-bonding layer **140b** includes a bonding dielectric **141b** and at least one bonding pad **142b**. In some embodiments, the bonding pad **142b** is surrounded by and at least partially exposed through the bonding dielectric **141b**. In some embodiments, the bonding pad **142b** extends through the bonding dielectric **141b** to electrically connect to the interconnect structure **124a** formed in the dielectric **124**.

[0045] In some embodiments, the sub-bonding layer **140a** and the sub-bonding layer **140b** can be bonded through a hybrid bonding process so as to form the bonding layer **140**. In some embodiments, the bonding pad **142a** and the bonding pad **142b** are aligned, so as to facilitate the hybrid bonding process. During the hybrid bonding process, the initial bond occurs at the dielectric-to-dielectric interface at room temperature under atmospheric conditions. The copper metal-to-metal connection is formed via annealing and metal diffusion.

[0046] As a result, the chip **110** can be bonded to the chip **120**, thereby forming a stacked structure. Similarly, the PMIC die **130** can be bonded to the chip **120** by performing a hybrid bonding process. For example, the bonding layer **150** can be formed by bonding a sub-bonding layer **150a** disposed on the chip **120** and a sub-bonding layer **150b** disposed on the power management die **130**.

[0047] In some embodiments, the sub-bonding layer **150a** is disposed over the RDL **128** formed on the semiconductive layer **122**. The sub-bonding layer **150a** includes a bonding dielectric **151a** and at least one bonding pad **152a**. The bonding pad **152a** extends through the bonding dielectric **151a** to electrically connect to the metal lines in the RDL **128**.

[0048] The sub-bonding layer **150b** is disposed over the power management die **130**, and includes a bonding dielectric **151b** and at least one bonding pad **152b**. In some embodiments, the bonding pad **152b** is aligned with the bonding pad **152a**, so that after the hybrid bonding process, the bonding pad **152b** can be bonded to the bonding pad **152a**. As a result, voltage conversion circuits (not shown) formed in the semiconductive layer **132** and the dielectric **134** in the power management die **130** can be coupled to the 3D capacitor structure **C2** in the chip **120** through the bonding layer **150**, the RDL **128**, the via **126**, and the interconnect structure **124a**, and can be coupled to the 3D capacitor structure **C1** in the chip **110** through the bonding layer **150**, the RDL **128**, the via **126**, the interconnect structures **124a**, the bonding layer **140**, the RDL **118**, the via **116**, and the interconnect structures **114a**.

[0049] Since the power management die **130** can be coupled to the 3D capacitor structure **C1** and the 3D capacitor structure **C2** through interconnects extending along the thickness (i.e., along the stacking direction as shown by the arrow in FIG. **1**) of the semiconductor structure **100**, the routing distances between the power management die **130** and the capacitors can be shortened, therefore, the signal integrity as well as the power integrity can be improved. Furthermore, since the stacked structure of the chip **110** and the chip **120** allows to integrate more capacitors within a small area to increase capacitance density, the power efficiency of the power management die **130** can also be

improved.

[0050] In some embodiments, the 3D capacitor structure C1 can be a 3D metal insulator metal (MIM) capacitor, while such out-of-plane dimension can be advantageously used to increase effective MIM area and related capacitance density. In some embodiments, the 3D capacitor structure C1 in the present disclosure may have a very high density, for example, higher than about 1 $\mu\text{m}/\text{mm}^2$. In some embodiments, the 3D capacitor structure can be a cylinder capacitor.

[0051] As shown in FIG. 23, in some embodiments, each of the 3D capacitor structures C1 includes a bottom metal plate 908, a top metal plate 910 over the bottom metal plate 908, and a plurality of 3D capacitor unit cells 912 are formed between the bottom metal plate 908 and the top metal plate 910. In some embodiments, a distance D1 between the bottom metal plate 908 and the top metal plate 910 is in a range of from about 1 μm to about 2 μm , which is much thinner than active or passive device components that are formed with deep trenches.

[0052] The configuration of the 3D capacitor unit cells 912 can have a crown-type capacitive structure or a concave-type capacitive structure. As illustrated in FIG. 23, in the embodiment that each of the 3D capacitor unit cells 912 are formed in crown type, the 3D capacitor unit cell 912 includes a first conductor film 914 and a second conductor film 916 stacked between the bottom metal plate 908 and the top metal plate 910. In some embodiments, the first conductor film 914 includes a first portion 914A connected to the bottom metal plate 908, and a second portion 914B connected to the first portion 914A and extending toward the top metal plate 910 from the bottom metal plate 908. In some embodiments, the second conductor film 916 is disposed adjacent to the first conductor film 914 and connected to the top metal plate 910, and extending toward the bottom metal plate 908 from the top metal plate 910. In some embodiments, the second conductor film 916 is vertically interleaving with the second portion 914B of the first conductor film 914. For instance, as the cross-sectional view shown in the figure, the second conductor film 916 is located adjacent to the inner and outer sides of an accommodated space 904. This accommodated space 904 is rounded by the first conductor film 914. In some embodiments, the second portion 914B can have a cylindrical shape, and the 3D capacitor structures C1 can be a cylindrical-type of capacitor.

[0053] Moreover, the 3D capacitor unit cell 912 further includes a first insulating film 928 for isolating the first conductor film 914 and the second conductor film 916. In other words, the MIM feature of the 3D capacitor structure C1 is performed by the stack of the first conductor film 914, the first insulating film 928, and the second conductor film 916. As shown in FIG. 23, in some embodiments, a second insulating film 930 can optionally be utilized to fill the space between the second conductor film 916 and the top metal plate 910. In some embodiments, the first insulating film 928 and the second insulating film 930 are composed of high-k dielectric material. For example, the high-k dielectric material may contain at least one of the oxides of Lanthanum, Hafnium, and Zirconium.

[0054] As shown in FIG. 24, in other embodiments, the 3D capacitor unit cell 912 are formed in concave type, the 3D capacitor unit cell 912 also includes the first conductor film 914 and the second conductor film 916 stacked between the bottom metal plate 908 and the top metal plate 910. In such embodiments, the second conductor film 916 is also formed adjacent to the first conductor film 914 and connected to the top metal plate 910, and extending toward the bottom metal plate 908 from the top metal plate 910. Comparing to the crown type shown in FIG. 23, the second conductor film 916 is laterally surrounded by the second portion 914B of the first conductor film 914 in each of the concave type capacitor cells 912, instead of entirely adjacent to inner and outer sides of the accommodated space 904 rounded by the first conductor film 914.

[0055] FIGS. 25A and 25B are top view diagrams illustrating a plurality of 3D capacitor unit cells 912 (concave type) according to some embodiments of the present disclosure. As shown in FIG. 25A, in some embodiments, the plurality of 3D capacitor unit cells 912 can be arranged to be a rectangular array between the metal plates (i.e., the bottom metal plate 908 and the top metal plate 910 previously shown in FIGS. 2 and 3) of the capacitor from a top view perspective. Alternatively,

in other embodiments, as shown in FIG. 25B, the 3D capacitor unit cells **912** can be arranged to be a hexagonal array between the metal plates of the capacitor from a top view perspective. Generally, the hexagonal-array-arrangement may provide a higher density of the 3D capacitor unit cells **912**. The arrangement of the 3D capacitor unit cells **912** of the present disclosure is not limited to the embodiments shown in FIG. 25A and 25B, because the 3D capacitor unit cells **912** may be arranged to be any symmetrical shape depending on the requirement.

[0056] In some embodiments, with the 3D capacitor structure **C1** and **C2** utilizing the 3D capacitor unit cells **912**, a capacitance density of the chip **110** and a capacitance density of the chip **120** can be respectively greater than $1\ \mu\text{F}/\text{mm}^2$. By stacking the chip **110** and the chip **120**, the capacitance density of the semiconductor structure **100** can be even doubled. In some embodiments, to further improve the power efficiency of the power management die **130**, more number of capacitor chips can be stacked in the semiconductor structure **100**. For example, in some embodiments, when two capacitor chips are stacked, the capacitance density may reach $5\ \mu\text{F}/\text{mm}^2$ and the power efficiency can be up to 85%. In addition, in some embodiments, when six capacitor chips are stacked, the capacitance density may reach $15\ \mu\text{F}/\text{mm}^2$ and the power efficiency can be up to 90%, and so on.

[0057] FIG. 2 shows a flow chart of a method **M1** for manufacturing the semiconductor structure **100** according to one embodiment of the present disclosure. The method **M1** includes steps **S110** to **S160**, but is not limited by the performing order shown in FIG. 2. FIGS. 3A to 3K show cross-sectional view of one or more stages of the method **M1** for manufacturing the semiconductor structure **100** according to one embodiment of the present disclosure.

[0058] In steps **S110** and **S120**, a wafer **10** and a wafer **20** are received. In some embodiments, a plurality of first chips **110** are formed in the wafer **10**, and a plurality of second chips **120** are formed in the wafer **20**. As shown in FIG. 3A, the wafer **10** includes the dielectric **114**, and the semiconductive layer **112** over the dielectric **114**. The 3D capacitor structure **C1** and the interconnect structure **114a** are formed within the dielectric **114**. In some embodiments, the wafer **20** and the wafer **10** can be identical to each other. However, the present disclosure is not limited thereto. As shown in FIG. 3B, the wafer **20** includes the dielectric **124**, and the semiconductive layer **122** over the dielectric **124**. The 3D capacitor structure **C2** and the interconnect structure **124a** are formed within the dielectric **124**.

[0059] In step **S130**, a bonding is formed between the wafer **10** and the wafer **20**. In some embodiments, such bonding can be formed by forming the sub-bonding layer **140a** on the wafer **10** as shown in FIG. 3C, forming the sub-bonding layer **140b** on the wafer **20** as shown in FIG. 3D, and bonding the sub-bonding layer **140a** to the sub-bonding layer **140b** as shown in FIG. 3E.

[0060] In some embodiments, as shown in FIG. 3C, the vias **116** in the semiconductive layer **112** can be formed in a via last (VL) manner. In such case, before forming the sub-bonding layer **140a**, the semiconductive layer **112** can be thinned so as to form the via **116** extending through the semiconductive layer **112**. In some embodiments, to provide the supporting strength when forming the via **116**, a carrier **CR1** can be attached on the dielectric **114** in advance (i.e. before the formation of the via **116**). In some embodiments, the carrier **CR1** can be a glass carrier, and can be attached to the dielectric **114** through an adhesion layer (not shown) that is removable (e.g., by heating or laser). After the via **116** is formed, the RDL **118** can be formed over the semiconductive layer **112** and the via **116**, and then the sub-bonding layer **140a** can be formed on the RDL **118**.

[0061] For the wafer **20**, the sub-bonding layer **140b** is formed on the dielectric **124** as shown in FIG. 3D. In such case, since the semiconductive layer **122**, which can be a part of a wafer or a substrate, is able to provide the supporting strength for forming the sub-bonding layer **140b**, a carrier may not be necessary during the formation of the sub-bonding layer **140b**.

[0062] After the sub-bonding layer **140a** and the sub-bonding layer **140b** are formed, the sub-bonding layer **140a** and the sub-bonding layer **140b** can be bonded to form the bonding layer **140** by performing a hybrid bonding process so that the wafer **10** and the wafer **20** can be bonded as

shown in FIG. 3E in step S130.

[0063] In step S140, a power management wafer 30 is received as shown in FIG. 3F. In some embodiments, a plurality of power management dies 130 are formed in the power management wafer 30. Furthermore, in step S150, the power management die 130 is bonded over the wafer 20 by bonding the power management wafer 30 on the wafer 20. That is, the power management dies 130 can be stacked on the second chips 120 in a wafer-on-wafer (WoW) manner.

[0064] In some embodiments, to bond the power management wafer 30 on the wafer 20, the sub-bonding layer 150a over the wafer 20 and the sub-bonding layer 150b over the power management wafer 30 can be formed first, and then, the two sub-bonding layer 150a and 150b can be bonded by a hybrid bonding process so as to form the bonding layer 150 that bonds the wafer 20 and the power management wafer 30.

[0065] As shown in FIG. 3G, before the sub-bonding layer 150a is formed, the via 126 can be formed in the semiconductive layer 122, and the RDL 128 can be formed on the semiconductive layer 122 and the via 126. In some embodiments, to form the via 126 that extends through the semiconductive layer 122 with finer pitches and sizes, the semiconductive layer 122 may be ground to be thinned down before forming the via 126. After the RDL 128 is formed, the sub-bonding layer 150a is formed on the RDL 128.

[0066] Furthermore, as shown in FIG. 3H, the sub-bonding layer 150b is formed on the dielectric 134 of the power management wafer 30. Afterwards, the sub-bonding layer 150a is bonded to the sub-bonding layer 150b; as a result, the power management wafer 30 can be bonded over the wafer 20 as shown in FIG. 3I.

[0067] Furthermore, as shown in FIG. 3I, after the power management wafer 30 is bonded to the wafer 20, the carrier CR1 can be removed from the dielectric 114 of the wafer 10, and openings 114b can be formed on the surface of the dielectric 114 so that the conductive bumps 119 can be formed on the dielectric 114 and coupled to the interconnect structures 114a formed in the dielectric 114 as shown in FIG. 3J. However, the present disclosure is not limited thereto. In some embodiments, another passivation layer may be formed on the dielectric 114 so as to facilitate the formation of the conductive bumps 119 thereon.

[0068] In step S160, the stacked structure of the wafer 10, the wafer 20, and the power management wafer 30 can be sawn by performing a sawing process as shown in FIG. 3K, and the semiconductor structure 100 can be singulated from the stacked wafers. In such case, edges of the power management die 130 can be aligned with edges of the chip 110 and edges of the chip 120. In some embodiments, the wafer 10 can be attached to a dicing tape DP1, and the blade B1 can cut the stacked structure from the surface of the power management wafer 30. However, the present disclosure is not limited thereto. In some other embodiments, the back side of the power management wafer 30 can be attached to the dicing tape DP1, and the blade B1 can cut the stacked structure from the surface of the wafer 10.

[0069] In the semiconductor structure 100, the chip 110 and the chip 120 are stacked with the back side of the chip 110 facing the front side of the chip 120 (i.e., in a front-to-back manner), that is, the semiconductive layer 112 of the chip 110 is in proximity to the dielectric 124 of the chip 120. However, the present disclosure is not limited thereto.

[0070] FIG. 4 shows a semiconductor structure 200 according to another embodiment of the present disclosure. The semiconductor structure 200 and the semiconductor structure 100 have similar structures, however, the difference between these two semiconductor structures 100 and 200 is in that, in the semiconductor structure 200, the chip 110 and the chip 120 are stacked with the front side of the chip 110 facing the back side of the chip 120, that is, the dielectric 114 of the chip 110 is in proximity to the semiconductive layer 122 of the chip 120.

[0071] In some embodiments, the semiconductor structure 200 can also be manufactured by the method M1. However, the performing orders of steps S110 to S160 may not be the same as shown in FIG. 2. FIGS. 5A to 5H show cross-sectional views of one or more stages of the method M1 for

manufacturing the semiconductor structure **200** according to one embodiment of the present disclosure.

[0072] As shown in FIGS. 5A and 5B, the steps **S120** and **S140** can be performed first to receive the wafer **20** and the power management wafer **30**. Step **S150** can be performed to bond the power management wafer **30** over the wafer **20**. In some embodiments, the bonding layer **150** that bonds the power management wafer **30** and the wafer **20** can be formed by forming the sub-bonding layer **150a** on the dielectric **124** of the wafer **20** as shown in FIG. 5C, forming the sub-bonding layer **150b** on the dielectric **134** of the power management wafer **30** as shown in FIG. 5D, and bonding the sub-bonding layer **150a** and the sub-bonding layer **150b** as shown in FIG. 5E.

[0073] Furthermore, as shown in FIG. 5E, after the power management wafer **30** and the wafer **20** are bonded, the semiconductive layer **122** can be ground to be thinned down so that the via **126** can be formed therein. The RDL **128** can be formed on the semiconductive layer **122** and the via **126**, and the sub-bonding layer **140b** can be formed on the RDL **128**.

[0074] Steps **S110** and **S130** are performed. As shown in FIG. 5F, the step **S110** of receiving the wafer **10** is performed. In some embodiments, to form the bonding between the wafer **10** and the wafer **20**, the sub-bonding layer **150a** is further formed on the dielectric **114** of the wafer **10** as shown in FIG. 5F. As a result, the step **S130** can be performed by bonding the sub-bonding layer **150a** to the sub-bonding layer **150b** as shown in FIG. 5G, and the stacked structure including the wafer **10**, the wafer **20** and the power management wafer **30** is formed. Afterwards, the semiconductive layer **112** can be ground to be thinned down, the via **116** can be formed in the semiconductive layer **112**, the RDL **118** can be formed on the semiconductive layer **112** and the via **116**, and the conductive bumps **119** as well as the passivation layer **117** can be formed on the RDL **118** as shown in FIG. 5H. In step **S160**, die sawing process can be performed to saw the stacked wafers so that the semiconductor structure **200** can be singulated.

[0075] In the semiconductor structure **200**, the chip **110** and the chip **120** are stacked with the front side of the chip **110** facing the back side of the chip **120**, that is, the first dielectric **114** of the chip **110** is in proximity to the second semiconductive **122** of the chip **120**. However, the present disclosure is not limited thereto. In some other embodiments, the chip **110** and the chip **120** can be stacked in a front-to-front manner or a back-to-back manner.

[0076] FIG. 6 shows a semiconductor structure **300** according to another embodiment of the present disclosure. The semiconductor structure **300** and the semiconductor structure **100** have similar structures, however, the difference between these two semiconductor structures **100** and **300** is in that, in the semiconductor structure **300**, the chip **110** and the chip **120** are stacked with the front side of the chip **110** facing the front side of the chip **120**, that is, the dielectric **114** of the chip **110** is in proximity to the dielectric **124** of the chip **120**.

[0077] In some embodiments, the semiconductor structure **200** can also be manufactured by the method **M1**. FIGS. 7A to 7H show cross-sectional view of one or more stages of the method **M1** for manufacturing the semiconductor structure **300** according to one embodiment of the present disclosure.

[0078] As shown in FIGS. 7A and 7B, the step **S110** and **S120** can be performed to receive the wafer **11** and the wafer **21**. In some embodiments, the wafer **11** and the wafer **21** are similar to the wafer **10** and the wafer **20**. However, in the wafer **11** and the wafer **21**, the vias **116** and **126** are formed in a via middle (VM) manner. That is, the vias **116** and **126** can be formed after the front end of line (FEOL) process and before the back end of line (BEOL) process. Therefore, the vias **116** and **126** are already respectively formed in the semiconductive layers **112** and **122** of the wafer **11** and the wafer **21** before the stacking process.

[0079] Step **S130** is performed to form the bonding between the wafer **11** and the wafer **21**. In some embodiments, to form the bonding layer **140**, the sub-bonding layer **140a** can be formed on the dielectric **114** of the wafer **11** as shown in FIG. 7C, the sub-bonding layer **140b** can be formed on the dielectric **124** of the wafer **21** as shown in FIG. 7D. Afterwards, a hybrid bonding process

can be performed to bond the sub-bonding layer **140a** and the sub-bonding layer **140b** so as to form the bonding layer **140** as shown in FIG. 7E. In such case, the wafer **11** and the wafer **21** are stacked in a front-to-front manner, and the bonding layer **140** can contact the dielectric **114** and the dielectric **124**.

[0080] Also, as shown in FIG. 7E, after the wafer **11** and the wafer **21** are bonded, the semiconductive layer **122** can be ground to expose the via **126** formed in the semiconductive layer **122**, the RDL **128** can be formed on the semiconductive layer **122** and the via **126**, and the sub-bonding layer **150a** can be formed on the RDL **128**.

[0081] The power management wafer **30** is received as shown in FIG. 7F when step S**140** is performed. Furthermore, to facilitate the bonding between the power management wafer **30** and the wafer **21**, the sub-bonding layer **150b** is further formed on the dielectric **134** of the power management wafer **30** as shown in FIG. 7F.

[0082] Step S**150** is performed so that the power management wafer **30** including a plurality of power management dies **130** can be bonded over the wafer **21** as shown in FIG. 7G. As a result, the stacked structure including the wafer **11**, the wafer **21**, and the power management wafer **30** is formed. Afterwards, the semiconductive layer **112** of the wafer **11** is ground to expose the via **116**, the RDL **118** is formed on the semiconductive layer **112** and the via **116**, and the conductive bumps **119** as well as the passivation layer **117** are formed on the RDL **118** as shown in FIG. 7H. In step S**160**, die sawing process can be performed to dice the stacking wafers so that the semiconductor structure **300** can be singulated.

[0083] Although each of the semiconductor structures **100**, **200**, and **300** includes two capacitor chips, the present disclosure is not limited thereto. In some embodiments, more capacitor chips can be stacked in the semiconductor structure so as to further improve the power efficiency of the voltage conversion performed by the PMIC.

[0084] FIG. **8** shows a semiconductor structure **400** according to another embodiment of the present disclosure. The semiconductor structure **400** and the semiconductor structure **300** have similar structures, however, the difference between these two semiconductor structures **300** and **400** is in that the semiconductor structure **400** includes more stacked chips. As shown in FIG. **8**, in addition to the chip **110** and the chip **120**, the semiconductor structure **400** further includes chips **160** and **170**.

[0085] In some embodiments, the chip **110**, the chip **120**, the chip **160**, and the chip **170** can have the same structure. For example, the chip **160** includes a dielectric **164**, and a semiconductive layer **162** over the dielectric **164**. Also, at least one 3D capacitor structure C**3** and at least one interconnect structure **164a** are formed in the dielectric **164**. Similarly, the chip **170** includes a dielectric **174**, and a semiconductive layer **172** over the dielectric **174**. Also, at least one 3D capacitor structure C**4** and at least one interconnect structure **174a** are formed in the dielectric **174**.

[0086] Furthermore, the chip **120** can be bonded to the chip **160** with a bonding layer **150** formed by bonding a sub-bonding layer **150a** formed on the chip **120** to a sub-bonding layer **150b** formed on the chip **160**. The chip **160** can be bonded to the chip **170** with a bonding layer **180** formed by bonding a sub-bonding layer **180a** formed on the chip **160** to a sub-bonding layer **180b** formed on the chip **170**. In addition, the chip **170** can be bonded to the power management die **130** with a bonding layer **190** formed by bonding a sub-bonding layer **190a** formed on the chip **160** to a sub-bonding layer **190b** formed on the power management die **130**. It may be noted that in the semiconductor structure **400**, different chips can be stacked in a front-to-front manner or a back-to-back manner. For example, the chip **110** and the chip **120** can be bonded in a front-to-front manner, and the chip **120** and the chip **160** can be bonded in a back-to-back manner. That is, the first dielectric **114** of the chip **110** is in proximity to the dielectric **124** of the chip **120**, and the semiconductive layer **122** of the chip **120** is in proximity to the semiconductive layer **162** of the chip **160**. In some embodiments, the capacitor chips can be stacked in a manner of front-to-back, front-to-front, or back-to-back depending on the requirements.

[0087] In some embodiments, the semiconductor structure **400** can also be manufactured by the method **M1** by repeating some of the steps. FIGS. **9A** to **9D** show cross-sectional view of one or more stages of the method **M1** for manufacturing the semiconductor structure **400** according to one embodiment of the present disclosure.

[0088] As shown in FIG. **9A**, steps **S110** to **S130** can be performed so that the wafer **11** and the wafer **21** can be bonded. In some embodiments, the processes shown in FIGS. **7A** to **7D** can be adopted to form the structure shown in FIG. **9A**. As a result, the semiconductive layer **122** is grinded to be thinner so as to expose the via **126**. The RDL **128** is formed on the semiconductive layer **122** and the via **126**, and the sub-bonding layer **150a** is formed on the RDL **128**.

[0089] Steps **S110** to **S120** can be repeated again for receiving the wafer **61** with a plurality of chips **160** formed therein and the wafer **71** with a plurality of chips **170** formed therein. The step **S130** can also be repeated for bonding the wafer **61** and the wafer **71** as shown in FIG. **9B**. In some embodiments, after the wafer **71** and the wafer **61** are bonded, the semiconductive layer **162** is ground to be thinner so as to expose the via **166**. The RDL **168** is formed on the semiconductive layer **162** and the via **166**, and the sub-bonding layer **150b** is formed on the RDL **168**. Afterwards, the wafer **61** can be further bonded to the wafer **21** with the bonding layer **150** formed by bonding the sub-bonding layers **150a** and **150b**. As a result, the four wafers **11**, **21**, **61**, and **71** can be stacked as shown in FIG. **9C**.

[0090] After the four wafers **11**, **21**, **61**, and **71** are stacked, steps **S140** and **S150** can be performed. For example, the power management wafer **130** can be bonded to the fourth wafer **170** through the bonding layer **190** formed by bonding the sub-bonding layers **190a** and **190b** as shown in FIG. **9D**. Furthermore, as shown in FIG. **9D** the semiconductive layer **112** of the wafer **110** can be ground to expose the via **116**, the RDL **118** can be formed on the semiconductive layer **112** and the via **116**, and the conductive bumps **119** as well as the passivation layer **117** can be formed on the RDL **118**. In step **S160**, die sawing process can be performed to dice the stacking wafers so that the semiconductor structure **400** can be singulated.

[0091] In the processes shown in FIGS. **3A** to **3H** and FIGS. **5A** to **5H**, the wafer **10** and the wafer **20** can have the same structures that have no via formed in the semiconductive layers upon receipt, that is, the vias are formed during the process of wafer stacking (i.e., via last). Also, in the processes shown in FIGS. **7A** to **7H** and FIGS. **9A** to **9C**, the wafer **11**, the wafer **21**, the wafer **61**, and the wafer **71** can have the same structures that have vias formed in the semiconductive layers upon receipt (i.e., via middle). However, the present disclosure is not limited thereto. In some embodiments, the semiconductor structures **100** to **400** can be manufactured by wafers having different structures.

[0092] FIGS. **10A** to **10E** show cross-sectional view of one or more stages of the method **M1** for manufacturing the semiconductor structure **100** according to one embodiment of the present disclosure.

[0093] In step **S110**, the wafer **11** that have vias **116** formed in the semiconductive layer **112** is provided as shown in FIG. **10A**, and in step **S120**, the wafer **20** that have no via formed in the semiconductive layer **122** is provided as shown in FIG. **10B**. Furthermore, as shown in FIGS. **10A** and **10B**, the sub-bonding layer **140a** is formed on the dielectric **114**, and the sub-bonding layer **140b** is formed on the second dielectric **124**. In such case, step **S130** can be performed to form the bonding layer **140** by bonding the sub-bonding layer **140a** and the sub-bonding layer **140b** through a hybrid bonding process. As a result, the wafer **11** and the wafer **20** can be bonded as shown in FIG. **10C**.

[0094] Furthermore, as shown in FIG. **10C**, the semiconductive layer **122** is ground to expose the via **126**, and the RDL **128** is formed on the semiconductive layer **122** and the via **126**. In addition, the sub-bonding layer **150a** is further formed on the RDL **128**. In such case, when the power management wafer **30** is received in step **S140**, the sub-bonding layer **150b** can be formed on the power management wafer **30** so that, in step **S150**, the power management wafer **30** can be bonded

to the wafer **20** through the bonding layer **150** formed by bonding the sub-bonding layer **150a** and the sub-bonding layer **150b** as shown in FIG. **10D**.

[0095] After the stacked structure of the wafer **11**, the wafer **20**, and the power management wafer **30** is formed, the semiconductive layer **112** of the wafer **11** is ground to expose the via **116**, the RDL **118** is formed on the semiconductive layer **112** and the via **116**, and the conductive bumps **119** as well as the passivation layer **117** are formed on the RDL **118** as shown in FIG. **10E**. In step **S160**, a sawing process can be performed to dice the stacking wafers so that the semiconductor structure **300** can be singulated.

[0096] In some embodiments, in the semiconductor structures **100**, **200**, **300**, or **400**, the power management die **130** is bonded to the capacitor chip in a wafer on wafer (WoW) manner, however, the present disclosure is not limited thereto. FIG. **11** shows a semiconductor structure **500** according to one embodiment of the present disclosure.

[0097] The semiconductor structure **500** and the semiconductor structure **100** have similar structures. However, the semiconductor structure **500** includes three capacitor chips **110**, **120**, and **160**, and the power management die **130** is bonded to the capacitor chip **160** through conductive bumps **539** and bump pads **569**. That is, the power management die **130** is stacked on the capacitor chips **160** in a chip on wafer (CoW) manner.

[0098] As shown in FIG. **11**, the semiconductor structure **500** includes the chips **110**, **120**, **160**, and the power management die **130**. The chip **110** can be bonded to the chip **120** through the bonding layer **140**, and the chip **120** can be bonded to the chip **160** through the bonding layer **150**. Furthermore, the semiconductor structure **500** further includes a plurality of bump pads **569** formed on the RDL **168** of the chip **160**, and a plurality of conductive bumps **539** formed on the dielectric **134** of the power management die **130**. In some embodiments, the conductive bumps **539** can be micro bumps that are aligned correspondingly with the bump pads **569** so that the power management die **130** can be bonded to the chip **160** by bonding the conductive bumps **539** to the bump pads **569**. In addition, the semiconductor structure **500** further includes an underfill **580** surrounding the conductive bumps **539** and is disposed over the chip **160** so as to protect the bonding between the conductive bumps **539** and the bump pads **569**. In some embodiments, to further protect the semiconductor structure **500**, a molding layer **590** is disposed on the chip **160** and surrounding the power management die **130**, the underfill **580** and the conductive bumps **539**. However, the present disclosure is not limited thereto. In some other embodiments, the underfill **580** and the molding layer **590** can be replaced by adopting a molded underfill (MUF) layer that can fill the gaps between the conductive bumps **539** and mold the power management die **130** at the same time.

[0099] FIGS. **12A** to **12K** show cross-sectional view of one or more stages for manufacturing the semiconductor structure **500** according to one embodiment of the present disclosure. As shown in FIGS. **12A** and **12B**, the wafer **10** and the wafer **20** are received. In some embodiments, a plurality of chips **110** are formed in the wafer **10**, and a plurality of chips **120** are formed in the wafer **20**. In some embodiments, the wafer **10** and the wafer **20** can have the same structures and can be replicas.

[0100] Furthermore, to facilitate the bonding between the wafer **10** and the wafer **20**, the sub-bonding layer **140a** and the sub-bonding layer **140b** are formed on the wafers **10** and **20** respectively. In some embodiments, the carrier CR1 is temporarily attached to the dielectric **114** of the wafer **10** so that the via **116** can be formed in the semiconductive layer **112** with enough supporting strength. The RDL **118** can be formed on the via **116** and the semiconductive layer **112**, and the sub-bonding layer **140a** can be formed on the RDL **118** as shown in FIG. **12C**. In addition, as shown in FIG. **12D**, the sub-bonding layer **140b** can be formed on the dielectric **124** of the wafer **20** without using carriers.

[0101] As a result, the sub-bonding layer **140a** and the sub-bonding layer **140b** can be bonded by a hybrid bonding process so that the wafer **10** can be bonded to the wafer **20** as shown in FIG. **12E**.

In addition, as shown in FIG. 12E, after the wafer 20 is bonded to the wafer 10, the via 126 can be formed in the semiconductive layer 122 of the wafer 20, the RDL 128 can be formed on the via 126 and the semiconductive layer 122, and the sub-bonding layer 150a can be formed on the RDL 128 so as to facilitate the bonding with the next chip.

[0102] In FIG. 12F, another wafer 61 including a plurality of chips 160 is received. Also, to bond the wafer 61 to the wafer 20, the sub-bonding layer 150b is formed on the dielectric 164 of the wafer 61. As a result, the sub-bonding layer 150a and the sub-bonding layer 150b can be bonded by a hybrid bonding process so that the wafer 61 can be bonded to the wafer 20 as shown in FIG. 12G.

[0103] After the wafer 61 is bonded to the wafer 20, the carrier CR1 can be removed, and openings can be formed on the surface of the dielectric 114 so that the conductive bumps 119 can be formed on the dielectric 114 and coupled to the interconnect structure 114a of the dielectric 114 as shown in FIG. 12H. An adhesive layer AD1 can be applied to the wafer 10 so as to protect the conductive bumps 119 and adhere a carrier CR2 to the stacked structure. In such case, the carrier CR2 is able to provide the supporting strength for the following processes, including grinding the semiconductive layer 162 to expose the via 166 therein, forming the RDL 168 on the semiconductive layer 162 and the via 166, and forming the bump pads 569 on the RDL 168. In some embodiments, the carrier CR2 can be a glass substrate.

[0104] In addition, a plurality of power management dies 130 can be received, and the plurality of conductive bumps 539 can be formed on the dielectric 134 of the power management dies 130. The power management dies 130 can be bonded over the wafer 61 by bonding the conductive bumps 539 to the bump pads 569 formed on the wafer 61 as shown in FIG. 12J. Furthermore, in some embodiments, after the power management dies 130 are bonded to the wafer 61, the underfill 580 can be applied to protect the connection between the conductive bumps 539 to the bump pads 569, and a molding process can be performed to mold the power management dies 130 on the wafer 61 with the molding layer 590. The carrier CR2 can be removed, and die sawing process can be performed to singulate the semiconductor structure 500 from the CoW structure as shown in FIG. 12K. In some embodiments, the wafer 10 can be attached to a dicing tape DP1, and the blade B1 can cut the stacked structure from the molding layer 590 surrounding the power management die 130. However, the present disclosure is not limited thereto. In some other embodiments, the molding layer 590 and the back sides of the power management dies 130 can be attached to the dicing tape DP1, and the blade B1 can cut the stacked structure from the surface of the wafer 10.

[0105] FIG. 13 shows a semiconductor structure 600 according to one embodiment of the present disclosure. The semiconductor structure 600 and the semiconductor structure 500 have similar structures. However, while the chips 110, 120, and 160 are all stacked in a front-to-back manner in the semiconductor structure 500, the chips 110, 120, and 160 can be stacked in different manners in the semiconductor structure 600, for example, the chip 120 can be stacked on the chip 110 in the front-to-front manner while the chip 160 can be stacked on the chip 120 in the front-to-back manners.

[0106] FIGS. 14A to 14H show cross-sectional view of one or more stages for manufacturing the semiconductor structure 600 according to one embodiment of the present disclosure. As shown in FIGS. 14A and 14B, the wafers 11 and 20 are received. In some embodiments, the wafer 11 may be different from the wafer 20 in that the wafer 11 has via formed in the semiconductive layer 112 while the wafer 20 does not.

[0107] Furthermore, to facilitate the bonding between the wafer 11 and the wafer 20, the sub-bonding layer 140a and the sub-bonding layer 140b are formed on the dielectrics 114 and 124 of the wafers 11 and 20 respectively. As a result, the sub-bonding layer 140a and the sub-bonding layer 140b can be bonded by a hybrid bonding process so that the wafer 11 can be bonded to the wafer 20 as shown in FIG. 14C. In addition, as shown in FIG. 14C, after the wafer 20 is bonded to the wafer 11, the via 126 can be formed in the semiconductive layer 122 of the wafer 20, the RDL 128 can be formed on the via 126 and the semiconductive layer 122, and the sub-bonding layer

150a can be formed on the RDL **128** so as to facilitate the bonding with the next chip.

[0108] In FIG. **14D**, another wafer **60** is received. Also, to bond the wafer **60** to the wafer **20**, the sub-bonding layer **150b** is formed on the dielectric **164** of the wafer **61**. As a result, the sub-bonding layer **150a** and the sub-bonding layer **150b** can be bonded by a hybrid bonding process so that the wafer **60** can be bonded to the wafer **20** as shown in FIG. **14E**.

[0109] After the wafer **60** is bonded to the wafer **20**, the semiconductive layer **162** can be ground to be thinner so as to form the via **166** in the semiconductive layer **162**. The RDL **168** can be formed on the semiconductive layer **162** and the via **166**, and then, the passivation layer **117** can be formed on the RDL **168** with openings for receiving the conductive bumps **119** later.

[0110] As shown in FIG. **14F**, after the conductive bumps **119** is formed, the adhesive layer AD1 can be applied to the wafer **60** so as to protect the conductive bumps **119** and adhere a carrier CR2 to the wafer stacking structure. In such case, the carrier CR2 is able to provide the supporting strength for the following processes, including grinding the semiconductive layer **112** to expose the via **116** therein, forming the RDL **118** on the semiconductive layer **112** and the via **116**, and forming the bump pads **519** on the RDL **118** as shown in FIG. **14G**.

[0111] A plurality of power management dies **130** can be received, and the plurality of conductive bumps **539** can be formed on the dielectric **134** of each power management die **130**. In such case, the power management dies **130** can be bonded over the wafer **60** by bonding the conductive bumps **539** to the bump pads **519** formed on the wafer **11** as shown in FIG. **14H**. Furthermore, in some embodiments, after the power management dies **130** are bonded to the wafer **11**, the underfill **580** can be applied to protect the connection between the conductive bumps **539** to the bump pads **519**, and a molding process can be performed to mold the power management dies **130** on the wafer **11** with the molding layer **590**. The carrier CR2 can be removed, and a sawing process can be performed to singulate the semiconductor structure **600** from the CoW structure.

[0112] In the processes shown in FIGS. **14A** to **14H**, the semiconductor structure **600** is manufactured by wafers having different structures, for example, the wafer **11** with via middle and the wafer **20** with via last are both adopted. However, the present disclosure is not limited thereto. In some embodiments, the semiconductor structure **600** can be manufactured by wafers of the same structures.

[0113] FIGS. **15A** to **15E** show cross-sectional view of one or more stages for manufacturing the semiconductor structure **600** according to another embodiment of the present disclosure. As shown in FIGS. **15A** and **15B**, the wafer **11** with a plurality of chips **110** formed therein and the wafer **21** with a plurality of chips **120** formed therein are received. In some embodiments, the wafer **11** and the wafer **21** can have the same structure that each has vias formed in the semiconductive layers **112** and **122** before the BEOL process.

[0114] Furthermore, to facilitate the bonding between the wafer **11** and the wafer **21**, the sub-bonding layer **140a** and the sub-bonding layer **140b** are formed on the dielectrics **114** and **124** of the wafers **11** and **21** respectively. As a result, the sub-bonding layer **140a** and the sub-bonding layer **140b** can be bonded by a hybrid bonding process so that the wafer **11** can be bonded to the wafer **21** as shown in FIG. **15C**. In addition, as shown in FIG. **15C**, after the wafer **21** is bonded to the wafer **11**, the semiconductive layer **122** of the wafer **21** can be ground to expose the via **126**, the RDL **128** can be formed on the via **126** and the semiconductive layer **122**, and the sub-bonding layer **150a** can be formed on the RDL **128** so as to facilitate the bonding with the next chip.

[0115] In FIG. **15D**, the wafer **61** with a plurality of chips **160** formed therein is received. In some embodiments, the wafer **61** can have the same structure as the wafer **11** and **21**. Also, to bond the wafer **61** to the wafer **21**, the sub-bonding layer **150b** is formed on the dielectric **164** of the wafer **61**. As a result, the sub-bonding layer **150a** and the sub-bonding layer **150b** can be bonded by a hybrid bonding process so that the wafer **61** can be bonded to the wafer **21** as shown in FIG. **15E**.

[0116] After the wafer **61** is bonded to the wafer **21**, the semiconductive layer **162** can be ground to be thinner so as to expose the via **166** formed in the semiconductive layer **162**. The RDL **168** can

be formed on the semiconductive layer **162** and the via **166**, and then, the passivation layer **117** can be formed on the RDL **168** with openings for receiving the conductive bumps **119** later. As a result, the structure shown in FIG. **15E** becomes the same as the structure shown in FIG. **14E**. Therefore, the following processes can also be referred to FIGS. **14F** to **14H**, and the redundant drawings are omitted for brevity. After the CoW structure that includes a plurality of semiconductor structures **600** is formed as shown in FIG. **14H**, a sawing process can be performed to singulate the semiconductor structure **600**.

[0117] In some embodiments, the semiconductor structure can include more capacitor chips so as to improve the power efficiency of voltage conversion. FIG. **16** shows a semiconductor structure **700** according to one embodiment of the present disclosure. As shown in FIG. **16**, the semiconductor structure **700** includes chips **110**, **120**, **160**, **170**, and a power management die **130**.

[0118] In some embodiments, the semiconductor structure **700** can also be manufactured by the method M1 by repeating some of the steps. FIGS. **17A** to **17E** show cross-sectional view of one or more stages of the method M1 for manufacturing the semiconductor structure **700** according to one embodiment of the present disclosure.

[0119] As shown in FIG. **17A**, steps S110 to S130 can be performed so that the wafer **11** and the wafer **21** can be bonded. In some embodiments, the processes shown in FIGS. **7A** to **7D** can be adopted to form the structure shown in FIG. **17A**.

[0120] Steps S110 to S120 can be repeated again for receiving the wafer **61** and the wafer **71**. The step S130 can also be repeated for bonding the wafer **61** and the wafer **71**. After the wafer **61** and the wafer **71** are bonded, the sub-bonding layer **150a** can be formed on the wafer **21**, and the sub-bonding layer **150b** can be formed on the wafer **61**. Afterwards, the wafer **61** can be further bonded to the wafer **21** with the bonding layer **150** formed by bonding the sub-bonding layers **150a** and **150b**. As a result, the four wafers **11**, **21**, **61**, and **71** can be stacked as shown in FIG. **17B**.

[0121] After the four wafers **11**, **21**, **61**, and **71** are stacked, the semiconductive layer **172** can be ground so as to expose the via **176**, and the RDL **178** can be formed on the semiconductive layer **172** and the via **176**. The passivation layer **117** can be formed on the RDL **178** with openings for receiving the conductive bumps **119** later as shown in FIG. **17B**.

[0122] After the conductive bumps **119** is formed on the passivation layer **117** and coupled to the RDL **178**, the adhesive layer AD1 can be applied to the wafer **71** so as to protect the conductive bumps **119** and adhere a carrier CR2 to the wafer stacking structure as shown in FIG. **17C**. In such case, the carrier CR2 is able to provide the supporting strength for the following processes, including grinding the semiconductive layer **112** to expose the via **116** therein, forming the RDL **118** on the semiconductive layer **112** and the via **116**, and forming the bump pads **519** on the RDL **118** as shown in FIG. **17D**.

[0123] In steps S140 and S150, a plurality of power management dies **130** can be received, and the plurality of conductive bumps **539** can be formed on the dielectric **134** of the power management dies **130**. In such case, the power management dies **130** can be bonded over the wafer **11** by bonding the conductive bumps **539** to the bump pads **519** on the wafer **11** as shown in FIG. **17E**. Furthermore, in some embodiments, after the power management dies **130** are bonded to the wafer **11**, the underfill **580** can be applied to protect the connection between the conductive bumps **539** to the bump pads **519**, and a molding process can be performed to mold the power management dies **130** on the wafer **11** with the molding layer **590**. Finally, the carrier CR2 can be removed, and a sawing process can be performed to singulate the semiconductor structure **700** from the CoW structure.

[0124] Since the semiconductor structures **100**, **200**, **300**, **400**, **500**, **600**, and **700** allow the power management die to be coupled to the capacitors in the capacitor dies through interconnects extending along the stacking direction, the routing distances between the power management die and the capacitors can be shortened, therefore, the signal integrity as well as the power integrity can be improved. Furthermore, since the stacking structure of the capacitor dies allows to integrate

more capacitance within a small area, the power efficiency of the power management die can also be improved.

[0125] In some embodiments, the semiconductor structures **100**, **200**, **300**, **400**, **500**, **600**, and **700** can be used to provide voltages for a system on chip (SoC). For example, each of the power management dies **130** in the semiconductor structures **100**, **200**, **300**, **400**, **500**, **600**, and **700** may include fully integrated voltage regulators (FIVRs) for the SoC. In some embodiments, the semiconductor structure **100**, **200**, **300**, **400**, **500**, **600**, or **700** be disposed in the same package with the SoC.

[0126] FIG. **18** shows a semiconductor device **80** according to one embodiment of the present disclosure. The semiconductor device **80** includes a printed circuit board (PCB) **801**, a substrate **802**, an SoC **803**, and a semiconductor structure **900**. In some embodiments, the semiconductor structure **900** can be any of the semiconductor structure **100**, **200**, **300**, **400**, **500**, **600**, and **700** or the like. In some embodiments, the substrate **802** can be a package substrate, and the SoC **803** can be disposed on a first surface **802a** of the substrate **802**. In addition, the semiconductor structure **900** is embedded in the substrate **802** with a plurality of conductive bumps **919** exposed from the first surface **802a** of the substrate **802** and coupled to the SoC **803**. That is, the semiconductor structure **900** is embedded in the substrate **802**, and can be packaged in the same package as the SoC **803** on the substrate **802**. In some embodiments, the substrate **802** further include a plurality of solder balls **8021** disposed on a second surface **802b** so that the substrate **802** can be soldered to the PCB **801**.

[0127] FIG. **19** shows a semiconductor device **81** according to one embodiment of the present disclosure. The semiconductor device **81** is similar to the semiconductor device **80**. However, the differences between the semiconductor device **80** and the semiconductor device **81** is in that the semiconductor structure **900** is disposed on the second surface **802b** of the substrate **802** in the semiconductor device **81**. That is, the semiconductor structure **900** is disposed on a landside of the substrate **802** in the semiconductor device **81**.

[0128] FIG. **20** shows a semiconductor device **82** according to one embodiment of the present disclosure. The semiconductor device **82** is similar to the semiconductor device **81**. However, the semiconductor device **82** is different from the semiconductor device **80** in that while the substrate **802** is disposed on a first surface **801a** of the PCB **801**, the semiconductor structure **900** is disposed on a second surface **801b** of the PCB **801** in the semiconductor device **82**. That is, the semiconductor structure **900** is disposed on a landside of the PCB **801** in the semiconductor device **82**.

[0129] FIG. **21** shows a semiconductor device **83** according to one embodiment of the present disclosure. The semiconductor device **83** is similar to the semiconductor device **80**. However, in the semiconductor device **83** a fan-out frame **804** including a copper pillars **8042**, a molding **8041**, a RDL **8043** and solder balls **8044** is disposed on the PCB **801**. Also, the semiconductor structure **900** is embedded in the molding **8041** and is coupled to the SoC **803** through some of the copper pillars **8042** and the RDL **8043**.

[0130] FIG. **22** shows a semiconductor device **84** according to one embodiment of the present disclosure. The semiconductor device **84** is similar to the semiconductor device **81**. However, the semiconductor device **84** is different from the semiconductor device **81** in that the semiconductor device **84** further includes an interposer **805** disposed on the substrate **802**, and the SoC **803** is disposed on a first surface **805a** of the interposer **805**. In addition, the interposer **805** includes a plurality of solder balls **8051** on a second surface **805b** for soldering to the substrate **802**. In such case, the semiconductor structure **900** can also be disposed on the second surface **805b** of the interposer **805**. That is, the semiconductor structure **900** is disposed on a landside of the interposer **805** in the semiconductor device **84**. In some embodiments, the semiconductor devices **80**, **81**, **82**, **83**, and **84** can also be seen as semiconductor structures that includes the semiconductor structure **900**.

[0131] In summary, the semiconductor devices and the method for manufacturing semiconductor devices provided by the embodiments of the present disclosure allow the power management die to be coupled to the capacitors in the capacitor dies through interconnects extending along the stacking direction, so the routing distances between the power management die and the capacitors can be shortened. As a result, the signal integrity as well as the power integrity can be improved. Furthermore, since the stacking structure of the capacitor dies allows to integrate more capacitance within a small area, the power efficiency of the power management die can also be improved.

[0132] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

[0133] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein, may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods and steps.

Claims

1. A semiconductor structure comprising: a first chip having a first 3-dimensional (3D) capacitor structure therein; a second chip having a second 3D capacitor structure therein; a first bonding layer bonding the first chip with the second chip; and a power management die bonded with the second chip, wherein the power management die is electrically connected to the first 3D capacitor structure and the second 3D capacitor structure through interconnects extending along a thickness of the semiconductor structure and across the first bonding layer.
2. The semiconductor structure of claim 1, wherein the power management die is bonded with the second chip by a second bonding layer, and from a top view, edges of the power management die are aligned with edges of the first chip and edges of the second chip.
3. The semiconductor structure of claim 1, wherein the first chip comprises a first dielectric, a first semiconductive layer over the first dielectric, a first via extending through the first semiconductive layer, a first RDL over the first semiconductive layer and the first via, and the first 3D capacitor structure is within the first dielectric.
4. The semiconductor structure of claim 3, wherein the first dielectric of the first chip is in proximity to a second dielectric of the second chip, the first semiconductive layer of the first chip is in proximity to the second dielectric of the second chip, or the first semiconductive layer of the first chip is in proximity to a second semiconductive layer of the second chip; and wherein the first bonding layer is in contact with the first RDL or in contact with the first dielectric.
5. The semiconductor structure of claim 1, wherein the power management die is bonded with the second chip by a plurality of first conductive bumps, and the semiconductor structure further comprising: an underfill disposed over the second chip and surrounding the plurality of first conductive bumps; and a molding layer disposed on the second chip and surrounding the power management die, the underfill, and the plurality of first conductive bumps.
6. The semiconductor structure of claim 1, wherein the first chip has a first surface and a second surface opposite to the first surface, the first chip further comprises a plurality of second conductive bumps on the first surface of the first chip, and the second surface of the first chip contacts the first

bonding layer.

7. The semiconductor structure of claim 1, wherein a capacitance density of the first chip and a capacitance density of the second chip are respectively greater than $1\text{ }\mu\text{F}/\text{mm}^2$.

8. The semiconductor structure of claim 1, wherein the first 3D capacitor structure comprises: a top metal plate; a bottom metal plate over the top metal plate; and a plurality of 3D capacitor unit cells formed between the top metal plate and the bottom metal plate.

9. The semiconductor structure of claim 8, wherein each of the 3D capacitor unit cells comprises: a first conductor film, comprising: a first portion connected to the bottom metal plate; and a second portion connected to the first portion and extending toward the top metal plate from the bottom metal plate; and a second conductor film adjacent to the first conductor film and connected to the top metal plate and extending toward the bottom metal plate from the top metal plate, wherein the second conductor film is vertically interleaving with the second portion of the first conductor film.

10. The semiconductor structure of claim 1, wherein the first 3D capacitor structure is a cylindrical-type capacitor.

11. The semiconductor structure of claim 1, further comprising: a printed circuit board (PCB); a substrate disposed on the PCB; and a system on chip (SoC) disposed on a first surface of the substrate and coupled to the first chip, the second chip, and the power management die.

12. The semiconductor structure of claim 11, wherein: the first chip, the second chip, and the power management die are embedded in the substrate with a plurality of conductive bumps exposed from the first surface of the substrate; or the first chip, the second chip, and the power management die are disposed on a second surface of the substrate.

13. A method for manufacturing a semiconductor structure, comprising: forming a bonding between a first wafer having a plurality of first 3-dimensional (3D) capacitor structures therein and a second wafer having a plurality of second 3D capacitor structures therein to bond the first wafer with the second wafer; bonding a plurality of power management dies over the second wafer; and performing a sawing process to form the semiconductor structure including a first chip, a second chip, a first bonding layer, one of the plurality of power management dies, one of the plurality of first 3D capacitor structures in the first chip and one of the plurality of second 3D capacitor structures in the second chip, wherein the one of the plurality of power management dies is electrically connected to the one of the plurality of first 3D capacitor structures and the one of the plurality of second 3D capacitor structures through interconnects extending along a thickness of the semiconductor structure and across the first bonding layer.

14. The method of claim 13, wherein the step of forming the bonding between the first wafer and the second wafer comprises forming a first sub-bonding layer over the first wafer, forming a second sub-bonding layer over the second wafer, and bonding the first sub-bonding layer with the second sub-bonding layer.

15. The method of claim 14, further comprising: receiving the first wafer comprising a first dielectric, a first semiconductive layer over the first dielectric, a first via extending through the first semiconductive layer, and the plurality of first 3D capacitor structures are within the first dielectric; attaching a carrier on the first dielectric of the first wafer; grinding the first semiconductive layer of the first wafer to expose the first via; and forming a first RDL on the first semiconductive layer and the first via of the first wafer; wherein the first sub-bonding layer is formed on the first RDL.

16. The method of claim 14, further comprising: receiving the first wafer comprising a first dielectric, a first semiconductive layer over the first dielectric, and the plurality of first 3D capacitor structures are within the first dielectric; attaching a carrier on the first dielectric of the first wafer; forming a first via in the first semiconductive layer; and forming a first RDL on the first semiconductive layer and the first via of the first wafer; wherein the first sub-bonding layer is formed on the first RDL.

17. The method of claim 14, further comprising: receiving the first wafer comprising a first dielectric, a first semiconductive layer over the first dielectric, and the plurality of first 3D

capacitor structures are within the first dielectric, wherein the first sub-bonding layer is formed on the first dielectric.

18. The method of claim 13, further comprising: receiving a power management wafer comprising the plurality of power management dies; wherein the step of bonding the plurality of power management dies over the second wafer comprises: forming a third sub-bonding layer over the second wafer; forming a fourth sub-bonding layer over the power management wafer; and bonding the third sub-bonding layer with the fourth sub-bonding layer.

19. The method of claim 13, wherein the each of the plurality of first 3D capacitor structures comprises a top metal plate, a bottom metal plate, and a plurality of 3D capacitor unit cells formed between the top metal plate and the bottom metal plate.

20. The method of claim 19, further comprising arranging the plurality of 3D capacitor unit cells as a rectangular array or a hexagonal array.
